

RealMan Robot rm_description User Manual V1.6

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Revision History

No.	Date	Comment
V1.0	2024-2-19	Draft
V1.1	2024-7-3	Amend(Add GEN72 adapter files)
V1.2	2024-9-11	Amend(Add ECO63 adapter files)
V1.3	2024-12-25	Amend(Add 63, 65, 75, ECO65 six-axis force adapter files and 63, 65, 75, ECO63, ECO65 integrated six-axis force adapter files)
V1.4	2025-4-7	Amend(Add GEN72_II adapter files)
V1.5	2025-11-13	Amend(Add RML63_III adapter files)
V1.6	2026-4-16	Amend(Add ECO62 and RX75 adapter files)

Content

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rm_description_Package_Description

rm_description is a function package for displaying the robot model and TF transformation. Through this package, we can realize the linkage effect between a virtual robotic arm in a computer and a real robot arm in reality. In the moveit2 control, we also need the support of this package. This package is introduced in detail in the following aspects.

- 1.Package use.
- 2.Package architecture description.
- 3.Package topic description.

Through the introduction of the three parts, it can help you:

- 1.Understand the package use.
- 2.Familiar with the file structure and function of the package.
- 3.Familiar with the topic related to the package for easy development and use.

Source code address:https://github.com/RealManRobot/ros2_rm_robot.git.

rm_description_Package_Use

First, after configuring the environment and completing the connection, we can directly start the node and run the rm_description package.

```
rm@rm-desktop:~$ ros2 launch rm_description rm_<arm_type>_display.launch.py
```

In practice, the above needs to be replaced by the actual model of the robotic arm. The available models are 65, 63, 63_III, 75, eco62, eco63, eco65, gen72, gen72_II, and rx75. For RX75, explicitly use rm_rx75_6fb_display.launch.py for RX75-6FB and rm_rx75_6fb_v_display.launch.py for RX75-6FB-V.

The command to start the six-axis force version is currently available for 63, 65, 75, and eco65:

```
rm@rm-desktop:~$ ros2 launch rm_description rm_<arm_type>_6f_display.launch.py
```

The command to start the integrated six-axis force version is currently available for 63, 63_III, 65, 75, eco63, eco65, and rx75:

```
rm@rm-desktop:~$ ros2 launch rm_description rm_<arm_type>_6fb_display.launch.py
```

The command to start the vision-enabled integrated six-axis force version is currently only available for rx75:

```
rm@rm-desktop:~$ ros2 launch rm_description rm_<arm_type>_6fb_v_display.launch.py
```

For example, the launch command of 65 robotic arm:

```
rm@rm-desktop:~$ ros2 launch rm_description rm_65_display.launch.py
```

The following screen appears in the interface after successful node startup.

```
[robot_state_publisher-1] [INFO] [1700623506.039934044] [robot_state_publisher]: got segment Link1
[robot_state_publisher-1] [INFO] [1700623506.040043372] [robot_state_publisher]: got segment Link2
[robot_state_publisher-1] [INFO] [1700623506.040067148] [robot_state_publisher]: got segment Link3
[robot_state_publisher-1] [INFO] [1700623506.040131638] [robot_state_publisher]: got segment Link4
[robot_state_publisher-1] [INFO] [1700623506.040145545] [robot_state_publisher]: got segment Link5
[robot_state_publisher-1] [INFO] [1700623506.040156925] [robot_state_publisher]: got segment Link6
[robot_state_publisher-1] [INFO] [1700623506.040167895] [robot_state_publisher]: got segment base_link
```

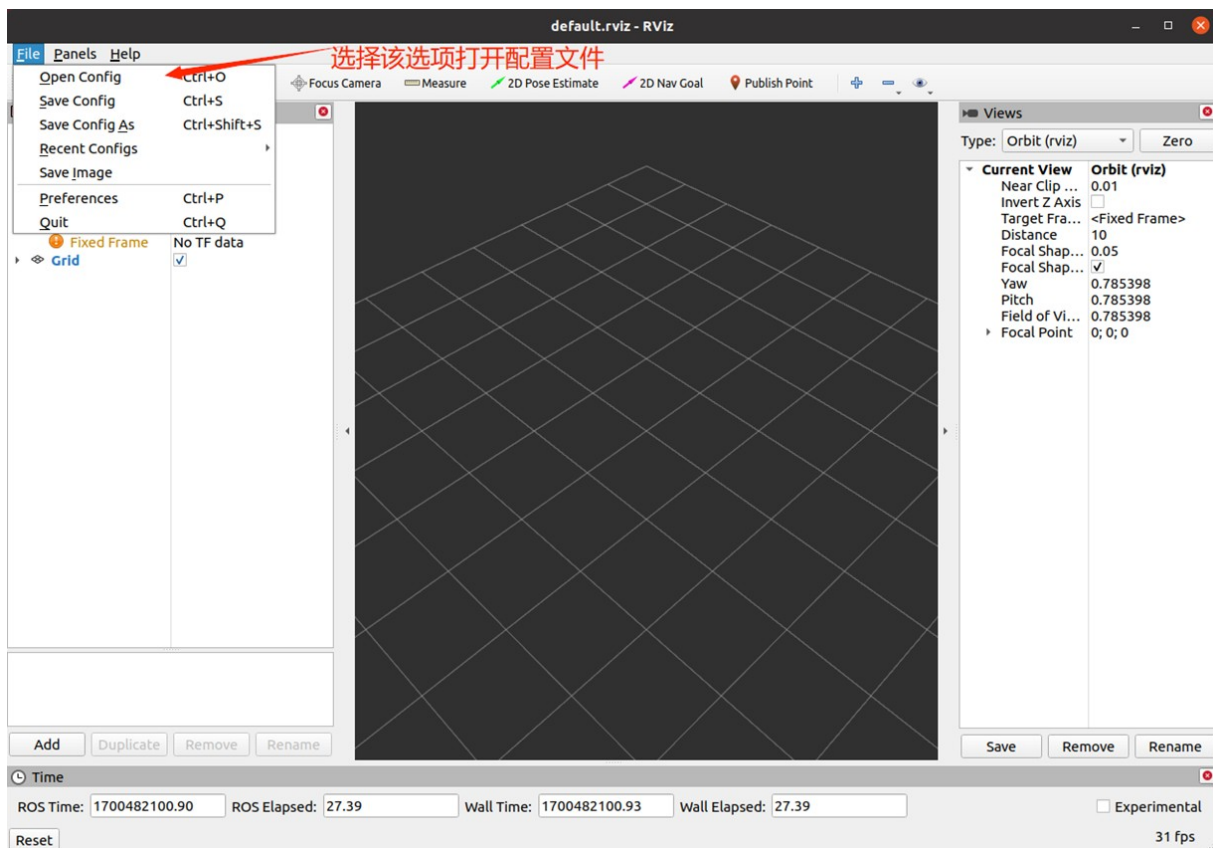
Then we need to launch the rm_driver node.

```
rm@rm-desktop:~$ ros2 launch rm_driver rm_<arm_type>_driver.launch.py
```

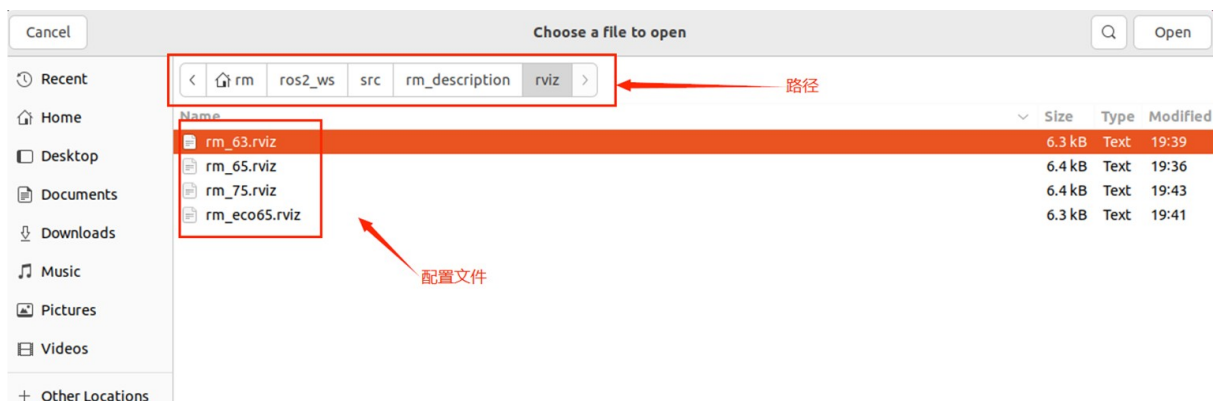
After a successful launch, we can check the state of the robotic arm in rviz2. Run the following command to launch rviz2.

```
rm@rm-desktop:~$ rviz2
```

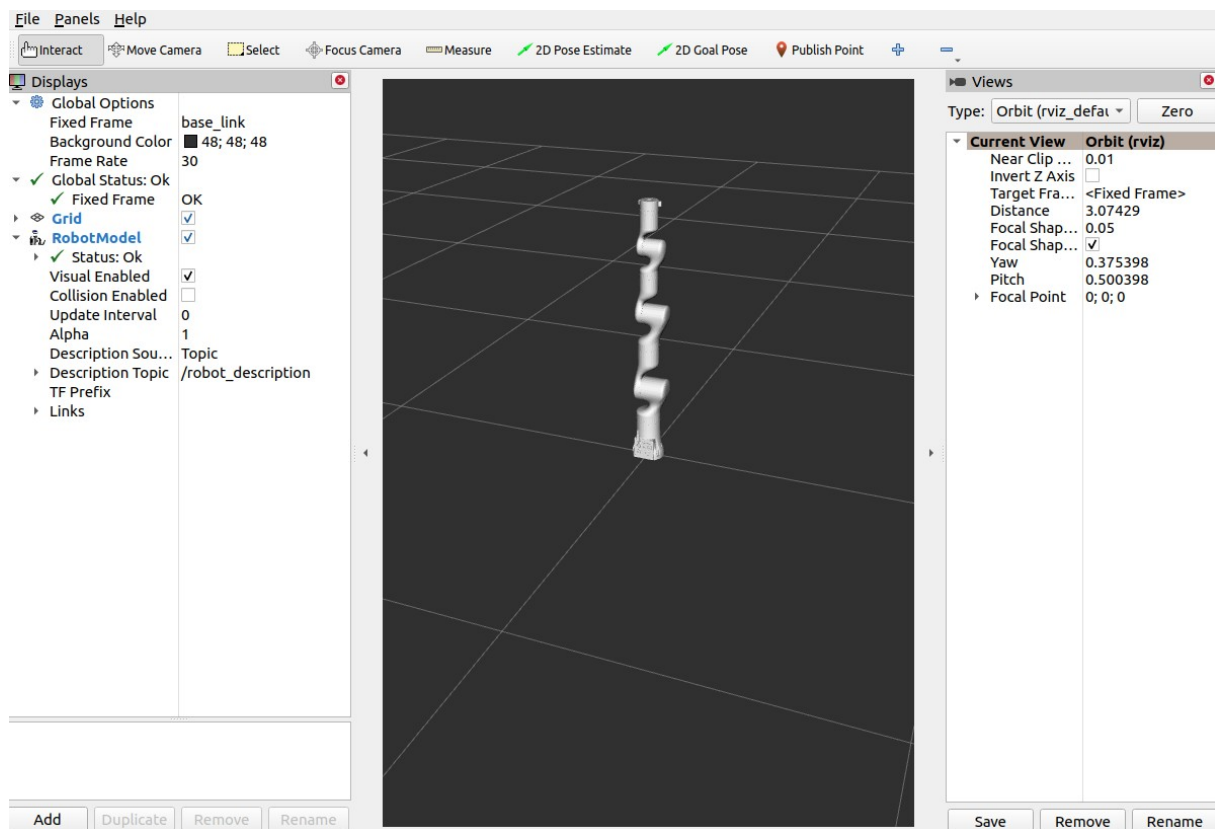
Open the robot model with the following configuration.



Find the corresponding configuration file under the rviz folder of the rm_description package.



After loading, you can see the current state of the robotic arm in the interface of rviz2.



rm_description_Package_Architecture_Description

Overview_of_package_files

The current rm_description package is composed of the following files.

```

CMakeLists.txt          # compilation rule file
launch
├── rm_63_6f_display.launch.py # 63 six-axis force launch file
├── rm_63_6fb_display.launch.py # 63 integrated six-axis force launch file
├── rm_63_display.launch.py    # 63 launch file
├── rm_63_III_6fb_display.launch.py # 63_III integrated six-axis force launch file
├── rm_63_III_display.launch.py # 63_III launch file
├── rm_65_6f_display.launch.py # 65 six-axis force launch file
├── rm_65_6fb_display.launch.py # 65 integrated six-axis force launch file
├── rm_65_display.launch.py    # 65 launch file
├── rm_75_6f_display.launch.py # 75 six-axis force launch file
├── rm_75_6fb_display.launch.py # 75 integrated six-axis force launch file
├── rm_75_display.launch.py    # 75 launch file
├── rm_eco62_display.launch.py # eco62 launch file
├── rm_eco65_6f_display.launch.py # eco65 six-axis force launch file
├── rm_eco65_6fb_display.launch.py # eco65 integrated six-axis force launch file
├── rm_eco65_display.launch.py # eco65 launch file
├── rm_eco63_6fb_display.launch.py # eco63 integrated six-axis force launch file
├── rm_eco63_display.launch.py # eco63 launch file
├── rm_gen72_II_display.launch.py # gen72_II launch file
├── rm_gen72_display.launch.py # gen72 launch file
├── rm_rx75_6fb_display.launch.py # RX75-6FB dual-arm launch file
├── rm_rx75_6fb_v_display.launch.py # RX75-6FB-V dual-arm launch file
└── meshes                      # model file storage folder
    ├── rm_63_arm              #63 robotic arm model file storage folder
    ├── base_link.STL
    ├── link1.STL
    ├── link2.STL
    ├── link3.STL
    ├── link4.STL
    ├── link5.STL
    ├── link6_6f.STL
    ├── link6_6fb.STL
    └── link6.STL

```

rm_65_arm	#65 robotic arm model file storage folder
— base_link.STL	
— link1.STL	
— link2.STL	
— link3.STL	
— link4.STL	
— link5.STL	
— link6_6f.STL	
— link6_6fb.STL	
— link6.STL	
rm_75_arm	#75 robotic arm model file storage folder
— base_link.STL	
— link1.STL	
— link2.STL	
— link3.STL	
— link4.STL	
— link5.STL	
— link6.STL	
— link7_6f.STL	
— link7_6fb.STL	
— link7.STL	
rm_eco62_arm	#eco62 robotic arm model file storage folder
— base_link.STL	
— Link1.STL	
— Link2.STL	
— Link3.STL	
— Link4.STL	
— Link5.STL	
— Link6.STL	
rm_eco65_arm	#eco65 robotic arm model file storage folder
— baseLink.STL	
— Link1.STL	
— Link2.STL	
— Link3.STL	
— Link4.STL	
— Link5.STL	
— Link6_6f.STL	
— Link6_6fb.STL	
— Link6.STL	
rm_eco63_arm	#eco63 robotic arm model file storage folder
— base_link.STL	
— Link1.STL	
— Link2.STL	
— Link3.STL	
— Link4.STL	
— Link5.STL	
— Link6_6fb.STL	
— Link6.STL	
rm_gen72_II_arm	#gen72_II robotic arm model file storage folder
— base_link.STL	
— Link1.STL	
— Link2.STL	
— Link3.STL	
— Link4.STL	
— Link5.STL	
— Link6.STL	
— Link7.STL	
rm_gen72_arm	#gen72 robotic arm model file storage folder
— base_link.STL	
— Link1.STL	
— Link2.STL	
— Link3.STL	
— Link4.STL	
— Link5.STL	
— Link6.STL	
— Link7.STL	
rm_rx75_arm	#RX75 dual-arm model file storage folder
— rm_rx75_left_arm	
— base_link.STL	

```

├── Link1.STL
├── Link2L.STL
├── Link3.STL
├── Link4.STL
├── Link5.STL
├── Link6.STL
├── Link7.STL
├── Link8.STL
├── rm_rx75_left_arm_v
│   ├── base_link.STL
│   ├── Link1.STL
│   ├── Link2.STL
│   ├── Link3.STL
│   ├── Link4.STL
│   ├── Link5.STL
│   ├── Link6.STL
│   ├── Link7.STL
│   └── Link8.STL
├── rm_rx75_right_arm
│   ├── base_link.STL
│   ├── Link1.STL
│   ├── Link2.STL
│   ├── Link3.STL
│   ├── Link4.STL
│   ├── Link5.STL
│   ├── Link6.STL
│   ├── Link7.STL
│   └── Link8.STL
├── rm_rx75_right_arm_v
│   ├── base_link.STL
│   ├── Link1.STL
│   ├── Link2.STL
│   ├── Link3.STL
│   ├── Link4.STL
│   ├── Link5.STL
│   ├── Link6.STL
│   ├── Link7.STL
│   └── Link8.STL
├── package.xml
├── README_CN.md
├── README.md
├── rviz                                     #rviz2 configuration file storage folder
│   ├── rm_63.rviz
│   ├── rm_65.rviz
│   ├── rm_75.rviz
│   ├── rm_eco62.rviz
│   ├── rm_eco65.rviz
│   ├── rm_eco63.rviz
│   ├── rm_gen72.rviz
│   └── rm_rx75.rviz
├── scripts
│   └── dual_arm_joint_state_bridge.py    # RX75 dual-arm joint state bridge
├── textures
├── urdf
│   ├── display_arm.rviz
│   ├── rm_eco62.csv
│   ├── rm_eco62_gazebo.urdf             #eco62 gazebo simulation urdf description file
│   ├── rm_eco62_gazebo.urdf.xacro       #eco62 gazebo simulation xacro description file
│   ├── rm_eco62.urdf                   #eco62 urdf description file
│   ├── rm_eco62.urdf.xacro              #eco62 xacro description file
│   ├── rm_65_6f.urdf                   #65 six-axis force urdf description file
│   └── rm_65_6fb.urdf                  #65 integrated six-axis force urdf description
file
│   ├── rm_65_description.csv
│   ├── rm_65_gazebo.urdf               #65 gazebo simulation urdf description file
│   ├── rm_65_gazebo.urdf.xacro         #65 gazebo simulation xacro description file
│   ├── rm_65.urdf                      #65 urdf description file
│   ├── rm_65.urdf.xacro                 #65 xacro description file
│   └── rm_75_6f.urdf                   #75 six-axis force urdf description file

```

```

file
├── rm_75_6fb.urdf                #75 integrated six-axis force urdf description
├── rm_75_description.csv
├── rm_75_gazebo.urdf            #75 gazebo simulation urdf description file
├── rm_75_gazebo.urdf.xacro      #75 gazebo simulation xacro description file
├── rm_75.urdf                   #75 urdf description file
├── rm_75.urdf.xacro             #75 xacro description file
├── rm_eco65_6f.urdf             #eco65 six-axis force urdf description file
├── rm_eco65_6fb.urdf           #eco65 integrated six-axis force urdf
description file
├── rm_eco65.csv
├── rm_eco65_gazebo.urdf         #eco65 gazebo simulation urdf description file
├── rm_eco65_gazebo.urdf.xacro   #eco65 gazebo simulation xacro description file
├── rm_eco65.urdf               #eco65 urdf description file
├── rm_eco65.urdf.xacro          #eco65 xacro description file
├── rm_eco63_6fb.urdf           #eco63 integrated six-axis force urdf
description file
├── rm_eco63.csv
├── rm_eco63_gazebo.urdf         #eco63 gazebo simulation urdf description file
├── rm_eco63_gazebo.urdf.xacro   #eco63 gazebo simulation xacro description file
├── rm_eco63.urdf               #eco63 urdf description file
├── rm_eco63.urdf.xacro          #eco63 xacro description file
├── rm_gen72.csv
├── rm_gen72_II.urdf            #gen72_II urdf description file
├── rm_gen72_II_gazebo.urdf      #gen72_II gazebo simulation urdf description
file
├── rm_gen72_gazebo.urdf         #gen72 gazebo simulation urdf description file
├── rm_gen72.urdf                #gen72 urdf description file
├── rm_rx75-6fb.urdf.xacro        #RX75-6FB dual-arm xacro description file
├── rm_rx75-6fb_v.urdf.xacro     #RX75-6FB-V dual-arm xacro description file
├── rm_rx75l-6fb.csv
├── rm_rx75l-6fb.urdf            #RX75-6FB left-arm urdf description file
├── rm_rx75l-6fb_v.csv
├── rm_rx75l-6fb_v.urdf          #RX75-6FB-V left-arm urdf description file
├── rm_rx75r-6fb.csv
├── rm_rx75r-6fb.urdf            #RX75-6FB right-arm urdf description file
├── rm_rx75r-6fb_v.csv
├── rm_rx75r-6fb_v.urdf          #RX75-6FB-V right-arm urdf description file
├── rml_63_6f.urdf               #63 six-axis force urdf description file
├── rml_63_6fb.urdf             #63 integrated six-axis force urdf description
file
├── rml_63_description.csv
├── rml_63_gazebo.urdf           #63 gazebo simulation urdf description file
├── rml_63_gazebo.urdf.xacro     #63 gazebo simulation xacro description file
├── rml_63.urdf                  #63 urdf description file
├── rml_63.urdf.xacro            #63 xacro description file

```

rm_description_Topic_Description

The following is the topic description of the package.

```

Subscribers:
  /joint_states: sensor_msgs/msg/JointState
  /parameter_events: rcl_interfaces/msg/ParameterEvent
Publishers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /robot_description: std_msgs/msg/String
  /rosout: rcl_interfaces/msg/Log
  /tf: tf2_msgs/msg/TFMessage
  /tf_static: tf2_msgs/msg/TFMessage
Service Servers:
  /robot_state_publisher/describe_parameters: rcl_interfaces/srv/DescribeParameters
  /robot_state_publisher/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
  /robot_state_publisher/get_parameters: rcl_interfaces/srv/GetParameters
  /robot_state_publisher/list_parameters: rcl_interfaces/srv/ListParameters
  /robot_state_publisher/set_parameters: rcl_interfaces/srv/SetParameters
  /robot_state_publisher/set_parameters_atomically:
rcl_interfaces/srv/SetParametersAtomically
Service Clients:

```

Action Servers:

Action Clients:

We mainly focus on the following topics. Subscribers: represents the topics it subscribes to, where `/joint_states` represents the current state of the robotic arm, which is published by our `rm_driver` package when running so that the model in `rviz` moves according to the actual state of the robotic arm. Publishers: represents the topics it currently publishes, where the most important published topics are `/tf` and `/tf_static`, which describe the coordinate transformation relationship between the joints of the robotic arm, namely TF transformation. There are relatively few remaining topics and service use scenarios and you can learn by yourself.